

THE BORDERED JACOBIAN OF BINARY-FORM MULTIPLICATION: AN INTEGRAL FOUNDATION FOR THE DEGREE-DIFFERENCE PRINCIPLE

ANONYMOUS

ABSTRACT. Let V_d denote the binary forms of degree d and let $m(A, B) = AB$ be multiplication $V_r \times V_s \rightarrow V_{r+s}$. Writing M for the Jacobian matrix of m in coefficient coordinates and $\kappa = (a_0, \dots, a_r, -b_0, \dots, -b_s)$ for the relative-scaling vector, we prove the *bordered Jacobian identity*: for every row vector v ,

$$\det \begin{pmatrix} M \\ v \end{pmatrix} = (-1)^{s(r+1)+1} \operatorname{Res}(A, B) \langle v, \kappa \rangle \quad \text{in } \mathbb{Z}[a, b, v],$$

valid for all $r, s \geq 1$ *including* $r = s$. The proof is elementary: interpolation at the roots plus one auxiliary point, a gap-Vandermonde identity, and the product formula. It uses no irreducibility or factoriality input, computes every sign, and, although it argues over $\mathbb{C}$, it delivers an integral identity valid after arbitrary base change.

Contracting the border against the Euler derivative of any bihomogeneous g of weight (p, q) gives $\det D(m, g) = (-1)^{s(r+1)+1} (p - q) g \operatorname{Res}$. The choice $g = \operatorname{Res}$ recovers the degree-difference identity $\det D\Phi_{r,s} = (-1)^{s(r+1)} (r - s) \operatorname{Res}^2$ of the July 2026 release, now over $\mathbb{Z}$; reduction shows $\Phi_{r,s}$ is étale on the coprime locus over any field in which $(s - r) \cdot 1$ is invertible. The choice $g = h \circ m$ gives zero: no function of the product can complete the chart. At $r = s$ the bordered identity does not degenerate; the collapse of $\Phi_{r,r}$ is purely the vanishing of the Euler weight $s - r$. In characteristic zero the normalisers are classified exactly: (m, g) is étale on the whole coprime locus if and only if $\delta g = c \operatorname{Res}^n$ for the scaling derivation δ , some constant $c \neq 0$ and $n \geq 1$ —equivalently, g is a constant multiple of a power of Res plus an element of $\ker \delta$; in characteristic p the same holds when $p \nmid n(s - r)$.

An accompanying exact suite verifies the identities of this paper per bidegree—the central identity in two separately written computational backends (SymPy and FLINT)—through $r + s \leq 8$ with spot checks to $(5, 6)$, with negative controls. We also give a formalization blueprint: a monic analogue of the bordered identity is already machine-checked inside Mathlib.

1. INTRODUCTION

The July 2026 release *The degree-difference principle and affine slices of binary-form factorisation spaces* [5] organised the geometry behind the three-dimensional Keller counterexample [1, 2] around one determinant identity. For binary forms A of degree r and B of degree s , with

$$\Phi_{r,s} : V_r \times V_s \longrightarrow V_{r+s} \times \mathbb{A}^1, \quad (A, B) \longmapsto (AB, \operatorname{Res}(A, B)),$$

it asserted

$$\det D\Phi_{r,s} = (-1)^{s(r+1)} (r - s) \operatorname{Res}(A, B)^2. \tag{1}$$

The proof given there ran through two fragile steps: the irreducibility of the resultant hypersurface combined with a unique-factorisation weight argument, to show the determinant is

Date: 6 August 2026 (v0.3-candidate).

2020 Mathematics Subject Classification. Primary 13P15; Secondary 14R15, 15A15, 15A54.

Key words and phrases. Jacobian conjecture, Keller map, binary forms, resultant, Sylvester matrix, Koszul complex, Vandermonde determinant, machine verification.

Version 0.3-candidate. Prepared for the Evidence Press evidence bundle; the verification suite, receipts, environment records, search reports and review correspondence ship with the release archive.

a constant multiple of Res^2 ; and a hand count of permutation inversions at the base point (X^r, Y^s) , to pin the constant. Its machine ledger checked the base-point sign matrix and the cubic-specific algebra, but not identity (1) itself at any bidegree.

This release replaces that foundation. The observation is that (1) is the shadow of a stronger, simpler, and integral statement about multiplication alone. Order the coefficient coordinates as $a_0, \dots, a_r, b_0, \dots, b_s$ and let M be the $(r+s+1) \times (r+s+2)$ Jacobian matrix of $m(A, B) = AB$. Let

$$\kappa = (a_0, \dots, a_r, -b_0, \dots, -b_s)$$

be the infinitesimal relative-scaling direction $(A, B) \mapsto (\lambda A, \lambda^{-1} B)$, which spans $\ker dm$ on the coprime locus.

Theorem 1.1 (Bordered Jacobian identity). *Let $r, s \geq 1$. For every row vector $v = (v_0, \dots, v_{r+s+1})$,*

$$\det \begin{pmatrix} M \\ v \end{pmatrix} = (-1)^{s(r+1)+1} \text{Res}(A, B) \langle v, \kappa \rangle$$

as an identity in $\mathbb{Z}[a_0, \dots, b_s, v_0, \dots, v_{r+s+1}]$. Equivalently, the signed maximal minors of M satisfy

$$(-1)^k \det M_{\hat{k}} = (-1)^{r(s+1)} \text{Res}(A, B) \kappa_k, \quad 0 \leq k \leq r+s+1,$$

where $M_{\hat{k}}$ deletes the k -th column. Both statements hold for all $r, s \geq 1$, including $r = s$.

The identity should be read as follows: *the top exterior power of the multiplication differential is the resultant times the contraction of the source volume form along the lost scaling direction.* Every Jacobian obtained by adjoining one scalar function to multiplication is then computed by a single pairing.

Corollary 1.2 (Contraction principle). *Let $g \in \mathbb{Z}[a_0, \dots, b_s]$ and write $\Phi_g = (m, g) : V_r \times V_s \rightarrow V_{r+s} \times \mathbb{A}^1$. Then*

$$\det D\Phi_g = (-1)^{s(r+1)+1} \text{Res}(A, B) \cdot dg(\kappa).$$

If g is bihomogeneous of weight (p, q) , meaning $g(\lambda A, \mu B) = \lambda^p \mu^q g(A, B)$, then $dg(\kappa) = (p-q)g$ and

$$\det D\Phi_g = (-1)^{s(r+1)+1} (p-q) g \text{Res}.$$

Three specialisations organise the whole subject.

- (1) **The degree-difference identity.** The resultant is bihomogeneous of weight (s, r) , so $g = \text{Res}$ gives $\det D\Phi_{r,s} = (-1)^{s(r+1)+1} (s-r) \text{Res}^2 = (-1)^{s(r+1)} (r-s) \text{Res}^2$: identity (1), now proved over $\mathbb{Z}$ (Corollary 4.1).
- (2) **The pullback obstruction.** If $g = h \circ m$ factors through the product, then $dg(\kappa) = dh(dm(\kappa)) = 0$, so $\det D\Phi_g = 0$ identically. No reparametrisation of the product alone can complete multiplication to a nondegenerate chart; the added coordinate must break the scaling symmetry (Corollary 4.3).
- (3) **The equal-degree collapse, localised.** Theorem 1.1 does not degenerate at $r = s$: the minors still equal $\pm \text{Res} \cdot \kappa_k$. What vanishes is the Euler weight $s - r$ of the resultant. The failure of $\Phi_{r,r}$ recorded in [5] is thus not a failure of the factorisation geometry; it is the statement that at equal degrees the resultant becomes invariant under the scaling torus, so its derivative can no longer see the lost direction (Remark 4.4).

Together (1)–(3) address, at the infinitesimal level, a question Tao raised when digesting the counterexample—he wrote that he did not have “a completely satisfactory geometric explanation for this miracle” [19]. Among candidate normalising functions, coefficient functionals pair with κ to single coefficients, which vanish on divisors meeting the coprime locus; functions

of the product pair to zero; the resultant pairs to $(s - r) \text{Res}$, which is nonvanishing on the *entire* coprime locus precisely when $r \neq s$. The resultant is not literally unique in this: in characteristic zero the normalisers are exactly the $g = c' \text{Res}^n + h$ with h in the kernel of the scaling derivation, and only the pullback part of that kernel is absorbed by target automorphisms (Proposition 4.5). The degree difference is the eigenvalue by which each power of the resultant reproduces itself under the scaling symmetry.

One limitation must be stated plainly. In its source context, Tao’s remark concerns the *global* phenomenon: why the repeated-root slice itself is affine three-space. The present identity explains the local étaleness and normalisation component of the construction; it does not explain that global fact, which rests on the slice geometry of [5].

What is gained over the July proof. The proof of Theorem 1.1 in Section 3 is elementary linear algebra: interpolation at the roots of A and B plus one auxiliary point, a gap-Vandermonde identity, the product formula for the resultant, and Zariski density. In particular:

- no irreducibility of the resultant hypersurface is used;
- no unique factorisation or normality is used;
- every sign is computed, and each signed constant is independently machine-verified per bidegree;
- the identity lives in $\mathbb{Z}[a, b]$, so reduction modulo p is legitimate: $\Phi_{r,s}$ is étale on the coprime locus over any field of characteristic p if and only if $p \nmid s - r$ (Corollary 4.2), sharpening the characteristic- p remark of [5] into a theorem.

Conceptually, Theorem 1.1 is the coordinate form of a classical principle: the resultant is the determinant of the degree- $(r + s)$ strand of the Koszul complex of (A, B) , a construction going back to Cayley and developed in modern form by Jouanolou, Chardin and Demazure [3, 10, 4, 7, 9]. The strand just above the Sylvester strand is exact with one-dimensional kernel spanned by κ , and its determinant is again the resultant. We have not found the bordered form with its two sign laws stated in the literature; the search boundary is recorded in Section 7. We make no claim of essential novelty for the principle itself, and every consequence drawn in Section 4 either strengthens or re-derives a statement of [5] on a firmer footing.

Machine verification and formalization. The accompanying suite `verify_bordered_jacobian.py` verifies each identity of this paper per bidegree, exactly over $\mathbb{Z}$ or $\mathbb{Q}$, in two separately written computational backends—SymPy symbolic determinants and FLINT integer-polynomial arithmetic with fraction-free elimination—together with negative controls that confirm the checks can fail (Section 5). The full determinant identity is verified for every bidegree with $r + s \leq 8$ and at the spot checks $(4, 5)$, $(5, 5)$, $(5, 6)$; the bordered/minor identity, including $r = s$, through $r + s \leq 7$; the base-point sign matrix for all $r, s \leq 40$.

The path to a complete machine-checked proof is shorter than one might expect: Mathlib’s `UniversalFactorizationRing` development already proves that the Jacobian matrix of the universal monic factorisation presentation *is* the Sylvester matrix up to sign—a square truncation of our matrix M —and that the universal coprime factorisation ring is étale over any base [17, 16]. The monic normalisation removes the scaling direction, so these results are an ingredient rather than an instance of Theorem 1.1; the degree-difference content itself remains to be formalized. Section 6 states the relationship precisely and lists the missing lemmas.

2. CONVENTIONS AND ELEMENTARY LEMMAS

Work over a commutative ring; all identities are in polynomial rings over $\mathbb{Z}$ and are proved by verification over $\mathbb{C}$ on a dense open set. Fix coordinates X, Y and write $A = \sum_{i=0}^r a_i X^{r-i} Y^i$,

$B = \sum_{j=0}^s b_j X^{s-j} Y^j$. Dehomogenising by $t = Y/X$ gives $a(t) = \sum a_i t^i$ and $b(t) = \sum b_j t^j$. The product $C = AB$ has coefficients $c_k = \sum_{i+j=k} a_i b_j$.

2.1. Sylvester convention. $\text{Res}(A, B)$ is the determinant of the $(r+s) \times (r+s)$ matrix whose rows are the coefficient vectors of $t^k a(t)$ for $0 \leq k < s$ followed by those of $t^k b(t)$ for $0 \leq k < r$, in the basis $1, t, \dots, t^{r+s-1}$. Then $\text{Res}(X^r, Y^s) = 1$, matching [5].

2.2. Jacobian data. M denotes the $(r+s+1) \times (r+s+2)$ matrix with rows indexed by $c_0, \dots, c_{r+s}$ and columns by $a_0, \dots, a_r, b_0, \dots, b_s$:

$$M[k, i] = \frac{\partial c_k}{\partial a_i} = b_{k-i}, \quad M[k, r+1+j] = \frac{\partial c_k}{\partial b_j} = a_{k-j},$$

with out-of-range subscripts read as zero. Column indices run from $k=0$ (the a_0 column) to $k=r+s+1$ (the b_s column).

2.3. Vandermonde notation. For points $x_1, \dots, x_m$ write $V(x_1, \dots, x_m) = \prod_{p < q} (x_q - x_p)$. For $0 \leq k \leq m$ let $W_k(x_1, \dots, x_m)$ be the $m \times m$ matrix whose rows are $(x_p^c)_{c \in \{0, \dots, m\} \setminus \{k\}}$.

Lemma 2.1 (Gap-Vandermonde). $\det W_k(x_1, \dots, x_m) = e_{m-k}(x_1, \dots, x_m) V(x_1, \dots, x_m)$, where e_j is the elementary symmetric polynomial.

Proof. Let T be an auxiliary variable and consider the full $(m+1) \times (m+1)$ Vandermonde matrix on the points $(x_1, \dots, x_m, T)$ with powers $0, \dots, m$. Its determinant is

$$V(x) \prod_p (T - x_p) = V(x) \sum_k (-1)^{m-k} e_{m-k}(x) T^k.$$

Expanding instead along the T -row gives $\sum_k (-1)^{m+k} T^k \det W_k(x)$. Comparing the coefficients of T^k gives the claim. $\square$

Lemma 2.2 (Product formula in this convention). Suppose $a(t) = a_r \prod_{i=1}^r (t - \alpha_i)$ and $b(t) = b_s \prod_{j=1}^s (t - \beta_j)$. Then

$$\text{Res}(A, B) = a_r^s b_s^r \prod_{i,j} (\beta_j - \alpha_i) = b_s^r \prod_j a(\beta_j) = (-1)^{rs} a_r^s \prod_i b(\alpha_i).$$

Proof. The transpose of the Sylvester matrix represents $\Psi : (u, w) \mapsto ua + wb$ on $V_{s-1} \times V_{r-1} \rightarrow V_{r+s-1}$ in monomial bases. Assume the α_i, β_j are pairwise distinct and $a_r b_s \neq 0$; the general case follows by density. Postcompose with evaluation E at the $r+s$ points $(\alpha_1, \dots, \alpha_r, \beta_1, \dots, \beta_s)$; $\det E = V(\alpha, \beta)$. In the evaluated basis, $(E\Psi)(u, w)$ has value $w(\alpha_i)b(\alpha_i)$ at α_i and $u(\beta_j)a(\beta_j)$ at β_j , so $E\Psi$ is block anti-diagonal: an $r \times r$ block $(b(\alpha_i)\alpha_i^c)_{c < r}$ against the w -columns and an $s \times s$ block $(a(\beta_j)\beta_j^c)_{c < s}$ against the u -columns. Hence

$$\det(E\Psi) = (-1)^{rs} \left(\prod_i b(\alpha_i) \right) V(\alpha) \left(\prod_j a(\beta_j) \right) V(\beta).$$

Dividing by $\det E = V(\alpha) V(\beta) \prod_{i,j} (\beta_j - \alpha_i)$ and substituting $\prod_i b(\alpha_i) = b_s^r \prod_{i,j} (\alpha_i - \beta_j)$ yields the displayed formulas. At $(A, B) = (X^r, Y^s)$ the matrix is the identity, confirming the normalisation. $\square$

Lemma 2.3 (Euler identity). $\text{Res}(\lambda A, \mu B) = \lambda^s \mu^r \text{Res}(A, B)$, and consequently $d \text{Res}(\kappa) = (s-r) \text{Res}$. More generally, if g is bihomogeneous of weight (p, q) then $dg(\kappa) = (p-q)g$.

Proof. Scaling a multiplies the s upper Sylvester rows by λ ; scaling b multiplies the r lower rows by μ . For the second claim, write $\delta = \sum_i a_i \partial_{a_i} - \sum_j b_j \partial_{b_j}$ for the *scaling derivation*, so that $dg(\kappa) = \delta g$ for every g by the definition of κ . On a monomial of bidegree (p, q) in the a - and b -variables, δ acts as multiplication by the integer $p - q$; hence $\delta g = (p - q)g$ for bihomogeneous g . This is a formal identity of polynomials, valid over any commutative ring. $\square$

Remark 2.4 (Worked example: $(r, s) = (1, 1)$). For $A = a_0X + a_1Y$, $B = b_0X + b_1Y$: the product has coefficients $c_0 = a_0b_0$, $c_1 = a_0b_1 + a_1b_0$, $c_2 = a_1b_1$; $R = a_0b_1 - a_1b_0$; $\kappa = (a_0, a_1, -b_0, -b_1)$; and, on the column order (a_0, a_1, b_0, b_1) ,

$$M = \begin{pmatrix} b_0 & 0 & a_0 & 0 \\ b_1 & b_0 & a_1 & a_0 \\ 0 & b_1 & 0 & a_1 \end{pmatrix}, \quad \sigma = (-1)^{r(s+1)} = +1.$$

Deleting the a_0 column: $\det M_{\hat{0}} = a_0R$, so $(-1)^0 \det M_{\hat{0}} = \text{Res} \cdot \kappa_0$. Deleting the b_1 column: $\det M_{\hat{3}} = b_1R$, so $(-1)^3 \det M_{\hat{3}} = -b_1R = \text{Res} \cdot \kappa_3$. Bordering by any v : $\det \begin{pmatrix} M \\ v \end{pmatrix} = -R(v_0a_0 + v_1a_1 - v_2b_0 - v_3b_1)$, the case $(-1)^{s(r+1)+1} = -1$ of [Theorem 1.1](#). Every sign convention of the paper can be audited on this display.

3. PROOF OF THE BORDERED JACOBIAN IDENTITY

By multilinearity of the determinant in the border row v , the bordered form of [Theorem 1.1](#) is equivalent to the minor form: expanding along the last row,

$$\det \begin{pmatrix} M \\ e_k \end{pmatrix} = (-1)^{(r+s+1)+k} \det M_{\hat{k}},$$

and $(-1)^{r+s+1} \cdot (-1)^{r(s+1)} = (-1)^{s(r+1)+1}$, so the two displayed sign laws match. It therefore suffices to prove, for each column index k ,

$$(-1)^k \det M_{\hat{k}} = (-1)^{r(s+1)} \text{Res}(A, B) \kappa_k. \quad (2)$$

Both sides are polynomials in $\mathbb{Z}[a, b]$, so it suffices to verify (2) on the dense open set $U \subset \mathbb{C}^{r+s+2}$ where $a_r b_s \neq 0$ and the combined list of roots $\alpha_1, \dots, \alpha_r$ of a and $\beta_1, \dots, \beta_s$ of b is pairwise distinct, within and across the two lists; U is a non-empty Zariski-open set. For the computation, an auxiliary point $\tau \in \mathbb{C}$ is chosen pointwise, distinct from all roots (such τ exist because $\mathbb{C}$ is infinite); τ enters only through the evaluation basis, every τ -dependent factor cancels, and the final expressions are independent of τ .

3.1. Evaluation. $M_{\hat{k}}$ is square of size $r + s + 1$. Change basis on the target (row space) by evaluating degree- $(r + s)$ polynomials at the $r + s + 1$ points $(\alpha_1, \dots, \alpha_r, \beta_1, \dots, \beta_s, \tau)$; the change-of-basis determinant is $V(\alpha, \beta, \tau)$. Writing V_E for the evaluation matrix, the rows of $V_E M_{\hat{k}}$ are, by the computation $\sum_k x^k \partial c_k / \partial a_i = x^i b(x)$ and $\sum_k x^k \partial c_k / \partial b_j = x^j a(x)$:

$$\begin{aligned} \text{row } \alpha_i : & \quad b(\alpha_i) \cdot (\alpha_i^c \mid 0), & c \text{ ranging over the surviving } a\text{-powers,} \\ \text{row } \beta_j : & \quad a(\beta_j) \cdot (0 \mid \beta_j^c), & c \text{ ranging over the surviving } b\text{-powers,} \\ \text{row } \tau : & \quad (b(\tau) \tau^c \mid a(\tau) \tau^c). \end{aligned}$$

3.2. Splitting the auxiliary row. Split the τ -row as $(b(\tau)\tau^c \mid 0) + (0 \mid a(\tau)\tau^c)$ and expand the determinant additively. In each of the two resulting matrices, count rows against columns: if the deleted column k lies in the b -block, the a -block retains $r + 1$ columns, which only the r rows α_i and possibly the τ -row can match; the term placing τ in the b -block therefore has $r + 1$ columns to fill with r rows and vanishes. Symmetrically, if k lies in the a -block, the term placing τ in the a -block vanishes. Exactly one term escapes this block-rank vanishing in

each case; at special points the surviving term may of course itself vanish, exactly when the corresponding coefficient does.

3.3. The b -block case. Let $k = r + 1 + j$ with $0 \leq j \leq s$, so the b -column of b_j is deleted. The surviving term is block-diagonal after moving the τ -row above the β -rows, a transposition past s rows contributing $(-1)^s$:

$$\det(V_E M_{\hat{k}}) = (-1)^s \left[\prod_i b(\alpha_i) \right] b(\tau) V(\alpha, \tau) \cdot \left[\prod_j a(\beta_j) \right] \det W_j(\beta),$$

where the a -block is the full Vandermonde on $(\alpha_1, \dots, \alpha_r, \tau)$ with powers $0, \dots, r$ and the b -block is the gap matrix of [Lemma 2.1](#). Divide by $V(\alpha, \beta, \tau) = V(\alpha) V(\beta) \prod_{i,j} (\beta_j - \alpha_i) \prod_i (\tau - \alpha_i) \prod_j (\tau - \beta_j)$ and cancel in three moves: $V(\alpha, \tau) = V(\alpha) \prod_i (\tau - \alpha_i)$ absorbs the α - τ factors; $b(\tau) = b_s \prod_j (\tau - \beta_j)$ absorbs the β - τ factors, leaving b_s ; and $\prod_i b(\alpha_i) = (-1)^{rs} b_s^r \prod_{i,j} (\beta_j - \alpha_i)$ absorbs the mixed factors, leaving $(-1)^{rs} b_s^r$. With $\det W_j(\beta) = e_{s-j}(\beta) V(\beta)$ ([Lemma 2.1](#)) this collects to

$$\det M_{\hat{k}} = (-1)^{s(r+1)} b_s e_{s-j}(\beta) \cdot b_s^r \prod_j a(\beta_j) = (-1)^{s(r+1)} b_s e_{s-j}(\beta) \text{ Res},$$

the last step by the second form of [Lemma 2.2](#). Since $b_j = (-1)^{s-j} b_s e_{s-j}(\beta)$, this reads $\det M_{\hat{k}} = (-1)^{s(r+1)} (-1)^{s-j} b_j \text{ Res}$, and hence

$$(-1)^k \det M_{\hat{k}} = (-1)^{r+1+j} (-1)^{s(r+1)+s-j} b_j \text{ Res} = (-1)^{r(s+1)} (-b_j) \text{ Res} = (-1)^{r(s+1)} \text{ Res } \kappa_k.$$

3.4. The a -block case. Let $k = i$ with $0 \leq i \leq r$. Now the term with the τ -row in the b -block survives, and the row order (α 's, β 's, τ) is already block-ordered, so no transposition sign appears:

$$\det(V_E M_{\hat{k}}) = \left[\prod_i b(\alpha_i) \right] \det W_i(\alpha) \cdot \left[\prod_j a(\beta_j) \right] a(\tau) V(\beta, \tau).$$

The same cancellations ($a(\tau) = a_r \prod_i (\tau - \alpha_i)$ now supplies the $\prod_i (\tau - \alpha_i)$) give

$$\det M_{\hat{k}} = (-1)^{rs} a_r e_{r-i}(\alpha) \text{ Res} = (-1)^{rs} (-1)^{r-i} a_i \text{ Res},$$

whence $(-1)^k \det M_{\hat{k}} = (-1)^{rs+r} a_i \text{ Res} = (-1)^{r(s+1)} \text{ Res } \kappa_k$, as $\kappa_i = +a_i$.

3.5. Conclusion. Identity (2) holds on the dense set U , hence in $\mathbb{Z}[a, b]$; nothing in the computation distinguished $r = s$. This proves [Theorem 1.1](#). $\square$

Remark 3.1. Let $S = \mathbb{Z}[X, Y]$ with its standard grading, and order the Koszul generators as (B, A) , so that the first differential in degree $r + s$ is $M : (\dot{A}, \dot{B}) \mapsto \dot{A}B + \dot{B}A$ and the second is κ , matching the sign convention $\kappa = (A, -B)$. The matrix $\begin{pmatrix} M \\ \kappa \end{pmatrix}$ is then the degree- $(r + s)$ strand of the Koszul complex of (A, B) , based and bordered: the strand $0 \rightarrow S_0 \xrightarrow{\kappa} S_r \oplus S_s \xrightarrow{M} S_{r+s} \rightarrow 0$ is exact on the coprime locus because the quotient $S/(A, B)$ by a coprime pair vanishes in degrees above $r + s - 2$, and [Theorem 1.1](#) says that the determinant of this based exact complex—not of any single differential—is the resultant, as it is for every strand of degree at least $r + s - 1$: the Cayley determinant [\[3, 10, 4, 7\]](#). The proof above is nothing but an explicit basing of that strand by interpolation. The bordering technique itself also has a classical antecedent: for $n - 1$ generic *linear* forms, Jouanolou proves that the signed maximal minors Δ_j of the coefficient matrix are the partial derivatives of the bordered determinant $\text{Res}(l_1, \dots, l_{n-1}, \lambda)$ and that $\text{Res}(l_1, \dots, l_{n-1}, f) = f(\Delta)$ [\[10, 5.4.1–5.4.4\]](#)—bordered evaluation at the Cramer point, with no resultant factor. Chardin records the scalar shadow of the minor form: the resultant divides every maximal minor of the degree- ν first Koszul differential for $\nu > \sum_i (d_i - 1)$, and is their gcd in the generic case [\[4,](#)

§IV, Remark 1]. Jouanolou’s formulary develops a bordered-determinant calculus for square elimination matrices—bordering a Macaulay-type matrix by a row u and a column v computes the apolarity functional, $\det \Theta_\nu^*(f; u, v) = c_\nu(f) \omega(\overline{uv})$ —and gradient-of-resultant identities modulo R for the square system, going back to [10, 4.6.3, 6.4.2–6.4.3]: the gradient of R with respect to distinguished coefficients reproduces the common zero mod R [12, 3.10.15, 3.10.25–3.10.31, 3.11.19.23–3.11.19.31]. The same corpus contains the square-system counterpart of a Jacobian–resultant identity: for n forms in n variables the Jacobian determinant with respect to the *variables* satisfies $\text{Jac}(f_1, \dots, f_n) \equiv d_1 \cdots d_n \det(f_{ij})$ modulo $(f_1, \dots, f_n)$, and the residue functional sends this twisted jacobian to the resultant [11, 3.8.1.9, 3.8.2.9]; the linear case of the signed-minor calculus, with $\Delta_i = \partial \Delta / \partial U_{1i}$ and a Rees-algebra structure modulo R , appears at [11, 3.5.5]. **Theorem 1.1** differs from all of these in differentiating with respect to the *coefficients* of a *factorisation* map, and in holding exactly over $\mathbb{Z}$ rather than modulo the ideal or through the residue pairing; there the natural constant is the *product* of the degrees, here it is their *difference*. Against Chardin’s divisibility statement in particular, it is the upgrade from “Res divides every maximal minor, and is their gcd generically” to an exact vector identity: each signed minor is identified as Res times the corresponding coordinate of the kernel vector, with both sign laws computed, and the Euler pairing then produces the degree difference. We prefer the bordered formulation because it computes, in one stroke, every Jacobian in which the missing coordinate might be sought.

4. CONSEQUENCES

Corollary 4.1 (Integral degree-difference identity). *In $\mathbb{Z}[a_0, \dots, b_s]$,*

$$\det D\Phi_{r,s} = (-1)^{s(r+1)}(r-s) \text{Res}(A, B)^2.$$

Proof. Border M with $v = d\text{Res}$ and contract: $\det D\Phi_{r,s} = (-1)^{s(r+1)+1} \text{Res} \cdot d\text{Res}(\kappa) = (-1)^{s(r+1)+1}(s-r) \text{Res}^2$ by **Lemma 2.3**. $\square$

Corollary 4.2 (Étaleness over fields and over $\mathbb{Z}[1/(s-r)]$). *Over any field F , the map $\Phi_{r,s}$ is étale at every point of the open subscheme $\{\text{Res} \neq 0\}$ (all scheme points) if and only if $(s-r) \cdot 1_F \neq 0$ in F : in characteristic zero, exactly when $r \neq s$; in characteristic $p > 0$, exactly when $p \nmid s-r$. For $r \neq s$, $\Phi_{r,s}$ is étale on the coprime locus over $\mathbb{Z}[1/(s-r)]$.*

Proof. The determinant is a unit times $(r-s) \text{Res}^2$ at such points. Conversely, if $(s-r) \cdot 1_F = 0$ the determinant vanishes identically over F . $\square$

Corollary 4.3 (Pullback obstruction). *If $g = h \circ m$ for any polynomial h on V_{r+s} , then $\det D(m, g) = 0$ identically. In particular no coefficient of the product, and no discriminant-type function of the product, can serve as the added coordinate: these fail identically, everywhere. By contrast, a source-coefficient functional such as a_0 fails only on a divisor (**Remark 4.4**).*

Remark 4.4 (Equal degrees). At $r = s$ the minor identity survives with sign $(-1)^{r(s+1)} = +1$: the *signed* maximal minors $(-1)^k \det M_{\hat{k}}$ equal $\text{Res} \cdot \kappa_k$ on the nose, and M has full rank $2r+1$ wherever $\text{Res} \neq 0$. The equal-degree collapse of (1) is therefore located entirely in **Lemma 2.3**: the resultant has weight $(s, r) = (r, r)$, is invariant under the scaling torus, and its Euler pairing with κ vanishes. More generally, by **Corollary 1.2** no bihomogeneous g of balanced weight can complete the chart at $r = s$, while an unbalanced g such as a single coefficient a_0 yields $\det D(m, a_0) = \pm a_0 \text{Res}$ —nondegenerate off a divisor, but never on the whole coprime locus, since every coefficient vanishes somewhere on it. The scaling direction at equal degrees is invisible to every function natural for the torus action; this is the precise infinitesimal content of the statement in [5] that $\Phi_{r,r}$ is everywhere degenerate.

Proposition 4.5 (Classification of normalisers). *Let F be a field, $S = F[a_0, \dots, a_r, b_0, \dots, b_s]$, $R = \text{Res}(A, B) \in S$, and let $\delta = \sum_i a_i \partial_{a_i} - \sum_j b_j \partial_{b_j}$ be the scaling derivation, so that $\det D(m, g) = (-1)^{s(r+1)+1} R \delta g$ for every $g \in S$ by [Theorem 1.1](#). Assume $r \neq s$.*

- (i) *(m, g) is étale at every point of the open subscheme $D(R) = \{\text{Res} \neq 0\}$ if and only if δg is a unit of the localisation $S[R^{-1}]$, i.e. $\delta g = cR^n$ with $c \in F^\times$ and $n \geq 1$.*
- (ii) *If $\text{char } F = 0$, the solutions are exactly*

$$g = \frac{c}{n(s-r)} R^n + h, \quad c \in F^\times, \quad n \geq 1, \quad h \in \ker \delta,$$

and among bihomogeneous g exactly the constant multiples $c'R^n$ with $c' \in F^\times$, $n \geq 1$.

- (iii) *If $\text{char } F = p > 0$, the same description holds for those n with $p \nmid n(s-r)$; the choice $g = cR^n$ fails whenever $p \mid n(s-r)$, and $\ker \delta$ is spanned by the monomials whose δ -weight is divisible by p , which in general strictly contains the invariants of the scaling torus.*

Proof. By [Corollary 1.2](#), (m, g) is étale throughout $D(R)$ if and only if $R \delta g$, equivalently δg , is a unit of $\Gamma(D(R), \mathcal{O}) = S[R^{-1}]$. Since S is a unique factorisation domain and R is irreducible over every field (geometric irreducibility over $\mathbb{Z}$: [\[10, 2.3\(iii\)\]](#)), the units of $S[R^{-1}]$ are $cR^\mathbb{Z}$ with $c \in F^\times$; as $\delta g \in S$, this gives $\delta g = cR^n$ with $n \geq 0$. The derivation δ preserves the weight decomposition $S = \bigoplus_w S_w$ and acts on S_w as multiplication by $w \cdot 1_F$; in particular the weight-zero component of δg vanishes, which excludes $n = 0$ and proves (i). Writing $g = \sum_w g_w$, the equation $\delta g = cR^n$ reads $w g_w = cR^n [w = n(s-r)]$; when $n(s-r)$ is invertible in F this determines $g_{n(s-r)} = cR^n / (n(s-r))$, leaves $g_0 \in \ker \delta$ free, and forces every other component to vanish—and when $p \mid n(s-r)$ there is no solution on that component, while $\delta(cR^n) = n(s-r)cR^n = 0$. A bihomogeneous g is concentrated in a single weight, which must then be $n(s-r) \neq 0$. The description of $\ker \delta$ in characteristic p is immediate from the action on monomials. $\square$

Remark 4.6 (The kernel is larger than the pullbacks). Only part of the ambiguity $h \in \ker \delta$ is geometrically trivial: for a pullback $h = h' \circ m$ the maps $(m, g + h)$ and (m, g) differ by the target automorphism $(C, z) \mapsto (C, z + h'(C))$, but $\ker \delta$ is strictly larger than the pullbacks. At $r = s = 1$, the element $h = a_0 b_1 - a_1 b_0$ satisfies $\delta h = 0$ and $h^2 = c_1^2 - 4c_0 c_2$ is a pullback, while h itself is not: it is antisymmetric under the swap $(A, B) \mapsto (B, A)$, which fixes every pullback. For such h , $(m, g + h)$ has the same Jacobian determinant as (m, g) without being obtained from it by a target automorphism. The same integer $s - r$ appears in the étaleness criterion, the $\mu_{|s-r|}$ -torsor, the class group $\mathbb{Z}/|s-r|\mathbb{Z}$ and the covering degree $|s-r| \binom{r+s}{r}$ of [\[5\]](#) because each of those statements consumes the determinant identity or the Euler weight as its local input; the global statements themselves are proved by the global arguments of [\[5\]](#), not by the pairing alone.

Remark 4.7 (Effect on the July release). All downstream results of [\[5\]](#) that consumed identity (1)—étaleness of the marked-root map, the torsor structure, the classification of linear-quadratic slices, and the Euler-characteristic exclusion of higher-degree untrimmed slices—now draw their *determinant input* from [Theorem 1.1](#) and [Lemma 2.3](#) alone; their global arguments are unchanged and remain as in [\[5\]](#). The dependency audit, statement by statement (each July item was reread against its text for this release):

July statement	determinant input consumed	status
Thm. 2.2 (identity)	—	superseded by Corollary 4.1
étaleness on $\{\text{Res} \neq 0\}$	the identity	re-proved over $\mathbb{Z}$ (Corollary 4.2)
Prop. 2.4 ($\mu_{ s-r }$ -torsor)	none (Kummer covers)	reread; independent
Prop. 2.5 (generic degree)	torsor count	reread; unchanged
Prop. 2.7, Cor. 2.8 (units, Cl, χ_c)	torsor + localisation	reread; unchanged
Prop. 3.3 (marked-root étale)	base change of identity	input replaced
§§4–6 (classification, tangent slice)	étaleness + slice geometry	reread; input via rows above

The two inputs the July proof of the identity needed (irreducibility of the resultant hypersurface; the base-point inversion count) are no longer load-bearing *for the identity*: irreducibility remains an input to the unit classification used in [Proposition 4.5](#) and in the global theory of [\[5\]](#), and the base-point computation survives as a machine-checked consistency anchor.

5. MACHINE VERIFICATION

The suite `verify_bordered_jacobian.py` accompanying this release verifies, exactly and with no floating point or sampling, the following checks. The backend of each check is stated. The two backends—SymPy symbolic and FLINT exact integer-polynomial with fraction-free elimination—are separately written construction and check paths, but they live in one suite, written by the same author from the same conventions: they guard against implementation error, not against a shared specification error, and their coefficientwise cross-comparison applies to the central determinant identity, not to every item. Receipts record package versions, platform and per-check timings; the degree bounds are chosen audit boundaries, not resource limits, except that (6, 7) and beyond were measured impractical (the (5, 6) spot check runs in about six minutes).

- (0) [Lemma 2.2](#) with symbolic roots, for $(r, s) \in \{(1, 1), (1, 2), (2, 2), (2, 3), (3, 3)\}$ [SymPy];
- (0') the gap-Vandermonde identity of [Lemma 2.1](#) for $2 \leq m \leq 5$, all k [SymPy];
- (I) [Corollary 4.1](#) for every bidegree with $r + s \leq 8$ and for (4, 5), (5, 5), (5, 6) [FLINT], and independently for $r + s \leq 5$ [SymPy], with coefficient-identical agreement of the two backends for $r + s \leq 5$;
- (II) the minor identity (2) with its sign law, for all bidegrees with $r + s \leq 7$ including $r = s$ [FLINT];
- (III) [Corollary 1.2](#) at four bidegrees for six weight cases, including three vanishing cases: balanced weight, a linear pullback, and a nonlinear pullback $g = c_0^2 + c_1$ [FLINT];
- (IV) the base-point sign matrix for all $1 \leq r, s \leq 40$ (1600 cases) [FLINT integer matrices];
- (V) the forward direction of [Corollary 4.2](#) via divisibility of every coefficient of $\det D\Phi_{r,s}$ by p for five pairs with $p \mid r - s$ [FLINT]; the converse direction is immediate from the identity and is not separately tested;
- (VI) negative controls: a perturbed Jacobian entry, a wrong sign, a wrong resultant power, a sign-flipped kernel vector, and transposed bidegrees are each detected as failures [FLINT].

The density step and the all-degree quantifier of [Theorem 1.1](#) are not machine-checkable by instance testing and are not claimed to be.

Two boundaries are declared explicitly. First, the per-bidegree checks certify instances, not the all-degree statement; the all-degree statement rests on the proof in [Section 3](#). Second, the ledger of the July release verified the base-point sign matrix and the cubic-specific algebra but not identity (1) itself; the present suite closes that gap and supersedes it. This claim about the July ledger is auditable against the archived release [\[5\]](#): its script (`verify_degree_`

`difference_affine_slices_final.py`, SHA-256 21dbdc38...¹) enumerates six checks in its receipt, none of which computes $\det D\Phi_{r,s}$ symbolically.

6. FORMALIZATION BLUEPRINT

Mathlib (master commit 1f0fbd1a, 6 August 2026) contains:

This paper	Mathlib analogue
Sylvester matrix, Res	<code>Polynomial.sylvester</code> , <code>Polynomial.resultant</code> [16]
Res unit $\Leftrightarrow$ coprime	<code>isUnit_resultant_iff_isCoprime</code>
Lemma 2.2	<code>resultant_eq_prod_eval</code> (monic normalisation)
Jacobian of monic multiplication is the Sylvester matrix (a square truncation of M ; see below)	<code>universalFactorizationMapPresentation_jacobiMatrix</code> [17]
Étaleness of monic coprime factorisation (related background; see below)	<code>UniversalCoprimeFactorizationRing</code> is étale [17]

The relationship between the last two rows and the present theorems must be stated with care, and is weaker than the word “chart” would suggest. The universal monic factorisation map fixes $a_r = b_s = 1$ and drops the (automatically monic) top coefficient of the product; its Jacobian is the square matrix obtained from M by deleting the a_r and b_s columns and the c_{r+s} row. Mathlib proves this square matrix is the Sylvester matrix, with determinant $\pm \text{Res}$, and that the coprime factorisation ring is étale over any base—*with no degree-difference factor*. Indeed the monic multiplication map is étale on its coprime locus even at $r = s$ (at $(r, s) = (1, 1)$ its Jacobian determinant is $b_0 - a_0 = -\text{Res}$), because fixing the leading coefficients removes the scaling direction that Theorem 1.1 and Corollary 4.2 are about. The Mathlib results are therefore an ingredient—the machine-checked determinant of the truncated matrix—rather than a monic instance of the present statements. What remains to formalize for the full Theorem 1.1: (i) bihomogeneity of the resultant (row-scaling of `Polynomial.sylvester`); (ii) the Euler identity Lemma 2.3; (iii) the bordered bookkeeping relating the non-monic $(r + s + 2)$ -column Jacobian, with its resultant row, to the truncated Sylvester block. All three are identities in polynomial rings over $\mathbb{Z}$, requiring no schemes and no analysis; (iii) is finite linear algebra over $\mathbb{Z}[a, b]$, and it—not the Mathlib material—carries the degree-difference content. The Keller disproof itself is already fully formalized: pull request #4474 to the Formal Conjectures repository, merged 26 July 2026 at commit 393aa9a [14]. A formalized Theorem 1.1 would therefore not fill a gap in the formal disproof; it would add a machine-checked *factorisation-space explanation* of the determinant that the disproof verifies directly.

7. PROVENANCE, PRIORITY AND SCOPE

The Keller counterexample was announced by Alpöge on 20 July 2026, crediting Fable for the construction [1, 2]; Gallagher exhibited an infinite family and Speyer a tangent-sweep geometric account in the same week, with expositions by Levinson and Tao [18, 13, 19]. Gao has since generalised the tangent-sweep mechanism to all dimensions ≥ 3 [8]; that framework is disjoint from the factorisation-space approach considered here. The determinant identity (1) was stated and proved for all bidegrees, over $\mathbb{C}$ and with the two inputs discussed above, in the July release [5]; the public threads contain the class-group observations of Skoof and Mondal cited there, but neither the identity nor its bordered refinement. The $(r, s) = (1, 2)$ case, with the same sign, appears as an explicit polynomial identity in Lou’s derivation note of

¹Full digest: 21dbdc3835302a40ed636e2090f42040d063a669fb1e37ede62d30be8f07834d.

20 July 2026 [15, Lemma 1]; the all-degree statement and the degree-difference factor are the July release’s. Adjacent determinantal literature on binary forms includes the Jacobian ideal of the binary discriminant [6].

The mathematical core of [Theorem 1.1](#)—the resultant as the determinant of any sufficiently high strand (degree at least $r + s - 1$) of the Koszul complex of two binary forms—is classical: Cayley’s 1848 construction, developed by Jouanolou and Chardin, with textbook treatments by Demazure and by Gelfand–Kapranov–Zelevinsky [3, 10, 4, 7, 9]. The full texts of [10], [11], [12] and [4] were audited for this release (6 August 2026). None contains a strand determinant above the Sylvester degree for two binary forms, the exact identification of the signed minors as $\text{Res} \cdot \kappa_k$, the sign laws, the Res^2 Jacobian, or the degree-difference factor. Their nearest contents, catalogued in [Remark 3.1](#), are: the linear-forms bordered-evaluation principle [10, 5.4.1–5.4.4]; gradient-of-resultant identities modulo R for the square elimination system [10, 4.6.3, 6.4.2–6.4.3], developed further in [12, 3.10.15, 3.10.31]; the bordered Macaulay-matrix calculus computing the apolarity functional [12, 3.10.25–3.10.30, 3.11.19.23–3.11.19.31]; determinantal-ideal computations for two binary forms *below* the Sylvester degree [12, 3.11.18.18–3.11.18.35]; the twisted-jacobian identities and the linear-case signed-minor calculus [11, 3.5.5, 3.8.1.9, 3.8.2.9]; and Chardin’s divisibility/gcd remark [4, §IV, Remark 1]. Within these audits and a bounded search of the remaining sources, the public discussion, and current literature databases, we did not find the bordered coordinate form with its sign laws $(-1)^{r(s+1)}$, $(-1)^{s(r+1)+1}$, the uniform treatment of $r = s$, or the contraction-principle derivation of (1). We therefore claim as contributions of this release: a complete elementary integral proof of [Theorem 1.1](#) with all signs computed; the derivation of the degree-difference identity, the characteristic- p criterion, the pullback obstruction and the equal-degree localisation from that single statement; the two-backend verification suite with negative controls; and the formalization blueprint. We claim no priority over the classical strand principle. The antecedent questions for all four primary classical sources — [10], [11], [12] and [4] — are settled by the full-text audits described above. The differential content of the corpus (jacobian and Fitting ideals of the resultant hypersurface [11, 3.5.3–3.5.4]; modulo- R gradient congruences [11, 3.7.1]; the twisted jacobian [11, 3.8]) is square-system material, differentiating with respect to the variables or working modulo R or through the residue pairing; it contains neither the bordered form nor the exact coefficient-side identity. One reading-scope note: within [10], §§4.6 and 6.4 are known to us through the corpus’s own citations ([11, 3.5.4, 3.7.1.4], [12, 3.10.15]) rather than by direct reading. The bounded-search qualification now applies only to the literature beyond this corpus.

Attribution. This manuscript, its proofs and its verification suite were produced by AI systems (Anthropic Claude, with the July release produced jointly with OpenAI Codex) operating under human direction, as part of the Evidence Press programme. The status of this release is: unrefereed candidate, awaiting independent verification.

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